

IEAP Series 02 RStudio

Statistical Tests

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```
library(tidyverse)
```

```
Warning: package 'ggplot2' was built under R version 4.3.3
```

Group Repository

GitHub Repository: <https://github.com/jenniferhoumard/IEAP-Series02-RStudio>

1 Team Workflow

1.1 a) Repository and Report Structure

Jennifer created the shared public GitHub repository and set up the project structure on the `main` branch before the group started working. This included the master Quarto document, the section files, and the overall organization of the notebook.

The report was divided into **11 separate Quarto sub-documents** and 2 other branches to correct the others. Each sub-document matched one of the main sections of the assignment, which made it easy to see what had to be completed and to keep track of the work.

Headings and subheadings were also added to each section document before the analysis started. This gave each team member a clear structure to follow and showed what information, code, results, and conclusions needed to be included.

All 11 section documents were included in one master Quarto document. This allowed everyone to work on separate files while still producing one organized final report. A table of contents was also added to the master document so that sections and subsections could be found easily.

We also created **12 Git branches**, with one branch for each section document. The branch names matched the section files, making it easy to identify which part of the report was being worked on and to follow the progress of each section.

Section	File
01	01_git_workflow.qmd
02	02_statistical_test_concepts.qmd
03	03_treatment_over_time.qmd
04	04_snorers_fatter.qmd
05	05_snorers_drink_smoke.qmd
06	06_men_fatter.qmd
07	07_women_smoke_less.qmd
08	08_variable_correlations.qmd
09	09_challenges_lessons.qmd
10	10_sources_references.qmd
11	11_checklist.qmd
12	12_final_report_fixes.qmd

1.2 b) Task Distribution

The branch structure and division of the sections were discussed as a group before starting the work. The sections were then distributed between the three team members so that everyone had clear responsibilities and could work independently on different files.

Team Member	Assigned Sections
Jennifer Houmard	01, 04, 07, 12
Jeanne Le Roux	03, 06, 09, 11
Jiangbin Long	02, 05, 08, 10

1.3 c) Branch Workflow

For each section, we used the same workflow:

1. A branch was created from `main` for the section being worked on.
2. The assigned student worked on the matching `.qmd` file.
3. Several meaningful commits were made while completing the section.
4. Once the section was finished, the branch was published to GitHub.
5. A pull request was opened from the section branch into `main`.
6. The pull request stayed open until the section was fully completed.
7. After all sections were finished, the three team members reviewed all pull requests together with a checklist defined together.
8. If something needed to be changed, comments were added and the section was corrected.
9. Once everyone agreed that the section was complete, the pull request was merged into `main`.

This workflow kept the `main` branch clean and ensured that only completed and reviewed sections were added to the final report.

1.4 Challenges Encountered and Lessons Learned

1. Adapting to a parallel Git workflow:

One of our main challenges was moving from a linear way of working, where one final version of a file is saved, to a parallel workflow where several people can work at the same time. We learned that branches allow each team member to work independently without affecting the work of others.

We also learned the importance of following a clear order: return to `main`, pull the latest changes, and then create a new branch. Skipping these steps can lead to merge conflicts. At first, we considered using only three branches, one per team member, but we decided that one branch per section was clearer and made it easier to track the work.

2. Working safely in a shared repository:

At first, there was some concern about making mistakes and potentially affecting other team members' work. Through the project, we learned that Git is designed to manage these situations safely.

Pull requests were especially useful because they allowed us to review and discuss changes before merging them into `main`. If something was incorrect, it could be changed on the branch before affecting the final report. This made the collaborative workflow much safer and gave us more confidence when working in the shared repository.

3. Protecting the `main` branch:

To reduce the risk of accidental changes, we protected the `main` branch using GitHub branch rules. Direct merging into `main` was restricted, and pull requests required at least one approval before they could be merged.

We also enabled protection against accidental deletion of the main branch and required review comments to be resolved before merging. This meant that completed sections could not be added to the final report without first being reviewed by another team member.

This helped us keep `main` as a stable version of the project and reduced the risk of accidentally deleting or merging unfinished work.

4. Planning the project and adapting to unexpected changes:

Another challenge was preparing all the shared files before the team started working on separate branches. Initially we forgot to add the data folder to the main GitHub repository, so the team had already started working on their section branches before the datasets were available.

The data folder then had to be added to `main` and pushed to GitHub. After this, each team member had to fetch the latest changes from `origin` in GitHub Desktop so that their local repository included the shared data files.

This showed us the importance of preparing the repository structure and shared files before creating working branches. It also helped us understand how changes made to `main` need to be synchronized with each team member's local repository during collaborative work.

5. Managing R libraries across different sections:

Because each team member worked on separate section files, we initially loaded the required R libraries inside our own `.qmd` files. This worked while developing the sections independently, but once everything was combined into the final report, the same libraries were being loaded repeatedly.

We also noticed that some code only worked on the computer of the person who had already loaded the required packages in their R session. This caused problems when another team member tried to run or render the same section from a clean session.

To solve this, the common libraries were moved to a setup chunk in the master Quarto document. This allowed the packages to be loaded once at the beginning of the report instead of being repeated throughout multiple sections. It also made the final report more consistent and easier to reproduce.

2 Statistical Tests: Comparison of Medians or Means

2.1 Make Clear What Is a Statistical Test, and When to Use It

Reference: (Vu & Harrington, 2020)

2.1.a Mean and Median

Concept	Definition & Reference
Mean	The sum of all values divided by the total number of values. Ch. 1, 1.4.1 <i>Measures of center: mean and median</i> , p. 30
Median	The middle value when the observations are ordered from smallest to largest. Ch. 1, 1.4.1 <i>Measures of center: mean and median</i> , p. 30

2.1.b Variance and Standard Deviation

Concept	Definition & Reference
Variance	A measure of how spread out the values are around the mean. A large variance means the values are more spread out. Ch. 1, 1.4.2 <i>Measures of spread</i> , p. 31
Standard deviation	Square root of the variance. Another measure of spread. How far values tend to be from the mean, in the same units as the data. Ch. 1, 1.4.2 <i>Measures of spread</i> , p. 31

2.1.c Normal Distribution

Concept	Definition & Reference
Normal distribution	A symmetric, bell-shaped distribution where most values are close to the mean. A perfectly normal distribution, the mean, median, and mode are equal. Ch. 3, 3.3 <i>Normal distribution</i> , pp. 152–153

2.1.d Statistical Test

Concept	Definition & Reference
Statistical test	A method used to decide whether a difference or relationship in the sample is likely to be real or could just be due to random chance. usually starting with a null hypothesis (H_0). used when we want to determine whether an observed difference/association in a sample provides sufficient evidence of a corresponding effect in the population, while accounting for sampling variability. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214

2.1.e When to Use a Statistical Test

Concept	Definition & Reference
When to Use a Statistical Test	used when we want to determine whether an observed difference/association in a sample provides sufficient evidence of a corresponding effect in the population, while accounting for sampling variability. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214

2.1.f Parametric and Non-Parametric Tests

Concept	Definition & Reference
Parametric test	A statistical test used for normally distributed data and makes more assumptions about the data. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214. Ch. 5, pp. 238, 245, 248
Non-parametric test	A statistical test used for non-normally distributed data and makes fewer assumptions. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214. Ch. 5, pp. 238, 245, 248

2.1.g Assumptions of Parametric Tests

Concept	Definition & Reference
Assumptions of Parametric Tests	Assumptions: normally distributed data, independent observations, and for some group comparisons, similar variances between groups. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214. Ch. 5, pp. 238, 245, 248

2.1.h Assumptions of Non-Parametric Tests

Concept	Definition & Reference
Assumptions of Non-Parametric Tests	Assumptions: Not normally distributed data, appropriately measured data, independence of observations when comparing independent groups. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214. Ch. 5, pp. 238, 245, 248

2.1.i p-value

Concept	Definition & Reference
p-value	Tells us how unusual our result would be if there were actually no real effect or difference. A small p-value means the result would be unlikely under the null hypothesis. Ch. 4, 4.3.1, Step 4, p. 214

2.1.j Risk of Error When Using a Statistical Test

Concept	Definition & Reference
Risk of Error When Using a Statistical Test	When performing a statistical test, we evaluate a sample to draw conclusions about a population. Because we rely on probabilities, there is always a risk of making an incorrect decision regarding the null hypothesis. (Ch. 4, 4.3.4 <i>Decision errors</i> , pp. 220–221)

2.1.k Paired and Unpaired Tests

Concept	Definition & Reference
Paired test	Used when observations are related, such as measurements taken from the same participants before and after treatment. Ch. 5, 5.2 <i>Two-sample test for paired data</i> , pp. 244–246
Unpaired test	Used when the groups are independent, containing different observations or participants. Ch. 5, 5.3 <i>Two-sample test for independent data</i> , pp. 247–250

2.2 Effect of Treatment Over Time

The `PrePost` dataset contains measurements taken before and after a treatment training. The same participants were measured at both time points, allowing us to examine whether the measured outcome changed after the treatment. Because the before and after measurements come from the same participants, the observations are paired.

2.2.a Import data

```
# 1.Import the data
```

```
prepost <- read_csv("data/PrePost copy.csv")
```

```
Rows: 16 Columns: 2
```

```
— Column specification —————
```

```
Delimiter: ","
```

```
chr (1): time
```

```
dbl (1): perf
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View the first 6 rows to make sure it was imported correctly
```

```
head(prepost)
```

```
# A tibble: 6 × 2
```

```
perf time
```

```
<dbl> <chr>
```

```
1 125 Before
```

```
2 130 Before
```

```
3 132 Before
```

```
4 135 Before
```

```
5 136 Before
```

```
6 138 Before
```

2.2.b Check the structure of the data

8 measurements before rehabilitation and 8 after:

```
Before: 125 130 132 135 136 138 140 145
```

```
After: 127 132 133 136 139 141 145 148
```

```
# Display the variable names
```

```
names(prepost)
```

```
[1] "perf" "time"
```

```
# Check the structure and data types  
str(prepost)
```

```
spc_tbl_ [16 × 2] (S3: spec_tbl_df/tbl_df/tbl/data.frame)  
$ perf: num [1:16] 125 130 132 135 136 138 140 145 127 132 ...  
$ time: chr [1:16] "Before" "Before" "Before" "Before" ...  
- attr(*, "spec")=  
.. cols(  
..   perf = col_double(),  
..   time = col_character()  
.. )  
- attr(*, "problems")=<externalptr>
```

```
# Obtain a first summary of the values  
summary(prepost)
```

```
      perf      time  
Min.   :125.0  Length:16  
1st Qu.:132.0  Class  :character  
Median :136.0  Mode   :character  
Mean    :136.4  
3rd Qu.:140.2  
Max.    :148.0
```

2.2.c Variables of the dataset

Variable	Description	Unit	Possible Values	Expected Change
perf	Performance measurement taken before and after cardiac rehabilitation	Not specified	Numerical values; observed in this dataset from 125 to 148	Expected to increase if a higher performance score represents better physical capacity
time	Indicates whether the performance measurement was taken before or after rehabilitation	None	Before, After	Not applicable; this is the measurement occasion

2.2.d Data analysis

Q: Does the rehabilitation treatment have an effect on measured performance from before to after treatment?

a) What comparison are we making?

- performance before rehabilitation vs performance after rehabilitation
 - Before: 125 130 132 135 136 138 140 145
 - After: 127 132 133 136 139 141 145 148
- If the same 8 participants are being compared before and after then this is a paired comparison

b) Parametric or non-parametric test?

- For a paired test, what matters is the distribution of the differences:

- $difference = After - Before$
- The differences are [2 2 1 1 3 3 5 3]
- Check that the differences are normally distributed using a **Shapiro-Wilk normality test**.
 - If $p > 0.05$ = not statistically significant and there is not enough evidence to say the differences depart from normality (we do not reject normality) → **parametric test** → **paired t-test** (Kim, 2015).
 - If $p < 0.05$ = statistically significant and there is enough evidence to say the differences are not normally distributed (we reject normality) → **non-parametric test** → **Wilcoxon signed-rank test** (Whitley & Ball, 2002).
 - For this dataset $p = 0.283 > 0.05$ therefore we do not reject normality of the differences and a parametric paired t-test can be used for statistical analysis since the data are paired (Kim, 2015).
- The **paired t-test** gives $p = 0.001 < 0.05$ → there is a statistically significant difference between performance before and after rehabilitation.

```
# 1. Calculate the differences between before and after in the dataset

# a) Separate the before and after performance values
before <- prepost %>% filter(time == "Before") %>% pull(perf)
# only keep rows where time == "Before" and extract the perf column

after <- prepost %>% filter(time == "After") %>% pull(perf)
# only keep rows where time == "After" and extract the perf column

# b) Calculate the difference for each participant
difference <- after - before

difference
```

```
[1] 2 2 1 1 3 3 5 3
```

```
# 2. Check for normality of data using Shapiro-Wilk normality test

shapiro.test(difference)
```

Shapiro-Wilk normality test

```
data: difference
W = 0.89896, p-value = 0.2828
```

```
# 3. Paired t-test comparing performance before and after rehabilitation
t.test(after, before, paired = TRUE)
```

Paired t-test

```
data: after and before
t = 5.4006, df = 7, p-value = 0.001008
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.405392 3.594608
sample estimates:
mean difference
      2.5
```

c) Which graph illustrates the effect of rehabilitation treatment?

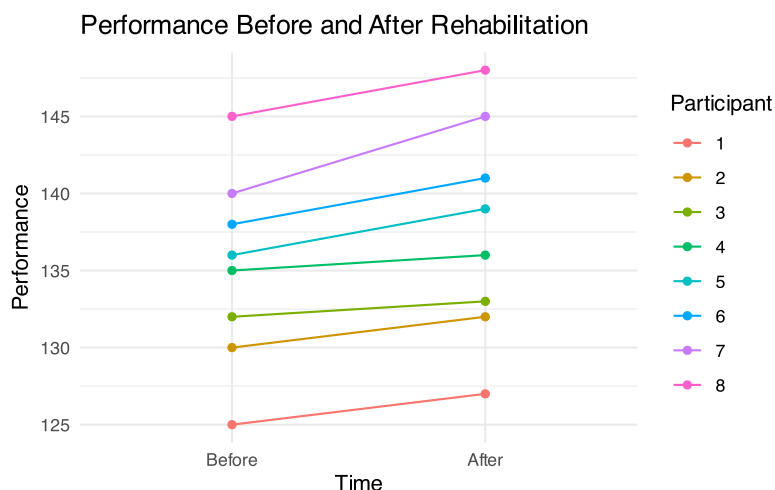
Since the same participants were measured before and after the rehabilitation treatment, a paired line plot was used. This type of graph connects each participant's before and after measurements, making it easy to see the direction and size of the change for each individual rather than only comparing the overall group averages.

```
# 1. Reorganize data structure so each participant has a id (number 1-8)

# a). Add a new column to dataset: participant IDs to label each participant
prepost_id <- prepost %>% mutate(id = rep(1:8, 2), time = factor(time, levels = c("Before",
"After"))) # give participants a # 1-8 and repeat twice. Before comes first and after comes second.

# 2. Create the paired line plot
ggplot(prepost_id, aes(x = time, y = perf, group = id, color = factor(id))) +
  #color = factor(id) → converts numbers to categorical data to label each person with a colour
  # group = id → connect the correct person's points

  geom_line() +
  geom_point() +
  labs(title = "Performance Before and After Rehabilitation",
       x = "Time", y = "Performance", color = "Participant") + # create legend labeled participant
  theme_minimal()
```



d) Rehabilitation treatment interpretation

Mean performance increased after rehabilitation compared with before. The slightly larger standard deviation after rehabilitation indicates that participant performance values were slightly more spread out around the mean.

- **Before rehabilitation:** 135.1 ± 6.2 = mean $\pm$ standard deviation
- **After rehabilitation:** 137.6 ± 7.0 = mean $\pm$ standard deviation
 - Smaller SD = participants values are more similar to each other
 - Larger SD = participant values are more spread out.

```
# 1. Calculate the mean & SD before rehabilitation

mean(before)
```

```
[1] 135.125
```

```
sd(before)
```

```
[1] 6.197638
```

```
# 2. Calculate the mean & SD after rehabilitation
```

```
mean(after)
```

```
[1] 137.625
```

```
sd(after)
```

```
[1] 7.008923
```

2.3 Testing Some Stereotypes

2.3.a Are Snorers Fatter?

The `snore` dataset contains information about participants including their height, weight, age, sex, alcohol consumption, smoking status, and whether they snore. To investigate whether snorers are “fatter,” BMI is calculated from height and weight and then compared between participants who snore and those who do not. Since the two groups contain different participants, they are treated as independent groups.

2.3.a.a Import Data

```
# 1.Import the data
```

```
snore <- read_delim("data/snore copy.csv") # delim separates the columns correctly
```

```
Rows: 100 Columns: 7
```

```
— Column specification —————
```

```
Delimiter: "\t"
```

```
chr (3): sex, snore, tabac
```

```
dbl (4): age, weight, height, alcool
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View the first 6 rows to make sure it was imported correctly
```

```
head(snore)
```

```
# A tibble: 6 × 7
```

	age	weight	height	alcool	sex	snore	tabac
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<chr>	<chr>
1	47	71	158	0	H	N	0
2	56	58	164	7	H	0	N
3	46	116	208	3	H	N	0
4	70	96	186	3	H	N	0
5	51	91	195	2	H	0	0
6	46	98	188	0	F	N	N

2.3.a.b Chosen Comparison Variable

The BMI of snorers and non-snorers can be compared to gives us a measure of body mass relative to height, so it is more appropriate than simply asking whether snorers weigh more.

$BMI_{snorers}$ vs $BMI_{non-snorers}$

$$BMI = \frac{weight(kg)}{height(m)^2}$$

```
# 1. Create BMI variable: create a new column BMI with the height m
snore <- snore %>% mutate(BMI = weight / (height / 100)^2) # mutate adds a new column BMI

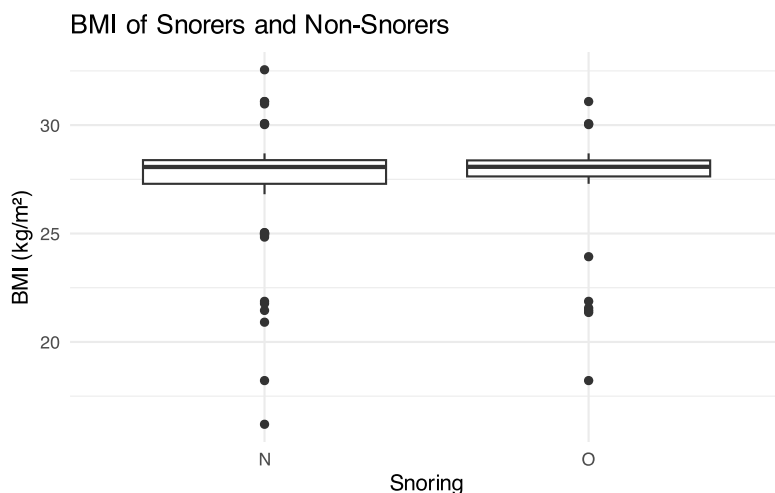
head(snore)
```

```
# A tibble: 6 × 8
  age weight height alcohol sex   snore tabac   BMI
<dbl> <dbl> <dbl>   <dbl> <chr> <chr> <chr> <dbl>
1    47    71   158     0 H     N     0    28.4
2    56    58   164     7 H     0     N    21.6
3    46   116   208     3 H     N     0    26.8
4    70    96   186     3 H     N     0    27.7
5    51    91   195     2 H     0     0    23.9
6    46    98   188     0 F     N     N    27.7
```

2.3.a.c Descriptive Analysis

```
# 2. # Visualize BMI distribution: boxplot before deciding the type of statistical test
```

```
ggplot(snore, aes(x = snore, y = BMI)) +
  geom_boxplot() +
  labs(title = "BMI of Snorers and Non-Snorers", x = "Snoring", y = "BMI (kg/m²)") +
  theme_minimal()
```



```
# 3. Descriptive statistics table
```

```
snore %>% group_by(snore) %>%
  summarise(mean_BMI = mean(BMI), median_BMI = median(BMI), sd_BMI = sd(BMI), n = n())
```

```
# A tibble: 2 × 5
  snore mean_BMI median_BMI sd_BMI    n
<chr>   <dbl>      <dbl> <dbl> <int>
1 N     28.4      28.4    4.5    6
2 O     28.5      28.5    4.5    6
```

1 N	27.3	28.1	2.81	65
2 0	27.1	28.1	2.83	35

Both groups are extremely similar and the boxplot also shows a very similar distribution and spread in both groups. Therefore descriptively, there is no obvious evidence that snorers have a higher BMI.

- Non-snorers (N): mean BMI = 27.27, median = 28.07, SD = 2.81, (n=65)
- Snorers (0): mean BMI = 27.11, median = 28.08, SD = 2.83, (n=35)

These are **unpaired/independent groups**, because the snorers and non-snorers are different people.

Before deciding the type of statistical test: check to see if BMI is normally distributed in each group using the Shapiro-Wilk test (Ghasemi & Zahediasl, 2012).

2.3.a.d Statistical Test Selection

```
# 4. Check the normality of BMI in both groups using Shapiro-Wilk test.
```

```
# a) Check normality of BMI for non-snorers
```

```
shapiro.test(snore$BMI[snore$snore == "N"]) # take the BMI column only for the rows where snore
= no
```

Shapiro-Wilk normality test

```
data: snore$BMI[snore$snore == "N"]
```

```
W = 0.78486, p-value = 2.323e-08
```

```
# b) Check normality of BMI for snorers
```

```
shapiro.test(snore$BMI[snore$snore == "0"]) # take the BMI column only for the rows where snore
= yes
```

Shapiro-Wilk normality test

```
data: snore$BMI[snore$snore == "0"]
```

```
W = 0.72636, p-value = 9.81e-07
```

BMI is not normally distributed in either group. Both are much smaller than (0.05), so we reject the assumption of normality for both groups:

- $BMI_{snorers} \rightarrow p = 9.81e - 07 < 0.05$
- $BMI_{non-snorers} \rightarrow p = 2.323e-08 < 0.05$

Since the two groups are independent and the data are non-normal, the **Wilcoxon rank-sum test** (also called the **Mann-Whitney U test**) was chosen (Whitley & Ball, 2002). This tests whether the BMI distributions differ between snorers and non-snorers.

2.3.a.e Statistical Comparison

```
# 5. Run the Wilcoxon rank-sum test on BMI
```

```
wilcox.test(BMI ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data: BMI by snore
W = 1128.5, p-value = 0.951
alternative hypothesis: true location shift is not equal to 0
```

$p = 0.951 > 0.05 \rightarrow$ there is no statistically significant difference in BMI between snorers and non-snorers. The BMI distributions of snorers and non-snorers are not meaningfully different in this sample.

2.3.a.f Conclusion

BMI was similar between snorers and non-snorers, with median values of 28.08 and 28.07 kg/m², respectively. The difference was not statistically significant (Wilcoxon rank-sum test, $W = 1128.5$, $p = 0.951$), providing no evidence to support the stereotype that snorers are fatter.

2.3.b Do Snorers Drink or Smoke More?

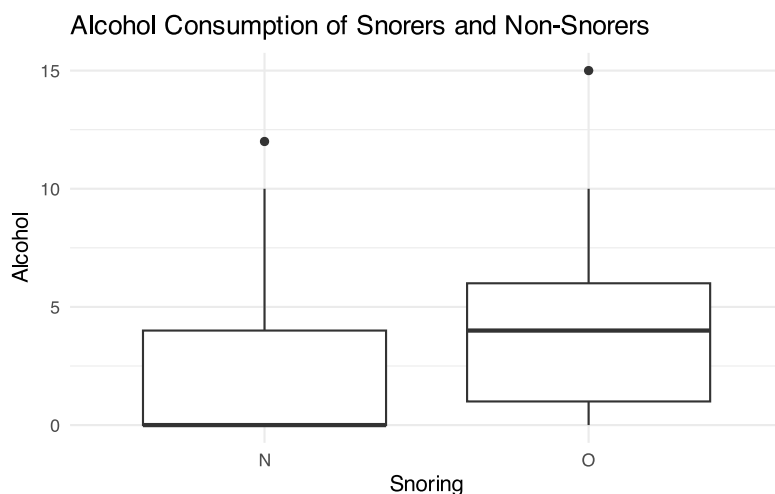
The `snore` dataset is used to compare alcohol consumption and smoking status between snorers and non-snorers. Alcohol is analyzed as a numerical variable, while smoking status is analyzed as a categorical variable.

Do snorers drink more?

Alcohol consumption has previously been associated with snoring and sleep-disordered breathing (Burgos-Sanchez et al., 2020). Therefore, alcohol consumption of snorers and non-snorers can be compared to determine whether snorers have higher alcohol consumption. Analyze the distribution of alcohol consumption before statistical analyses

$Alcohol_{snorers}$ vs $Alcohol_{non-snorers}$

```
# 1. Make a boxplot to compare alcohol consumption between snorers and non-snorers
ggplot(snore, aes(x = snore, y = alcool)) +
  geom_boxplot() +
  labs(title = "Alcohol Consumption of Snorers and Non-Snorers", x = "Snoring", y = "Alcohol") +
  theme_minimal()
```



```
# 2. Create a table to look at the descriptive statistics
```

```
snore %>% group_by(snore) %>%
  summarise(mean_alcool = mean(alcool), median_alcool = median(alcool), sd_alcool = sd(alcool),
n = n())
```

```
# A tibble: 2 × 5
  snore mean_alcool median_alcool sd_alcool    n
  <fct>    <dbl>         <dbl>    <dbl> <dbl>
1 N         2.50           2.50     2.50   10
2 O         4.00           4.00     2.50   10
```

	<chr>	<dbl>	<dbl>	<dbl>	<int>
1	N	2.37	0	3.18	65
2	0	4.03	4	3.47	35

Snorers appear to consume more alcohol than non-snorers. Snorers had a higher mean alcohol consumption (4.03 vs 2.37) and a higher median (4 vs 0). The boxplot also shows generally higher alcohol values among snorers.

- Non-snorers (N): mean alcohol = 2.37, median = 0, SD = 3.18
- Snorers (0): mean alcohol = 4.03, median = 4, SD = 3.47

```
# 3. Check the normality of alcohol consumption in both groups using the Shapiro-Wilk test

# a) Check normality for non-snorers
shapiro.test(snore$alcohol[snore$snore == "N"])
```

Shapiro-Wilk normality test

```
data:  snore$alcohol[snore$snore == "N"]
W = 0.76256, p-value = 6.818e-09
```

```
# b) Check normality for snorers
shapiro.test(snore$alcohol[snore$snore == "0"])
```

Shapiro-Wilk normality test

```
data:  snore$alcohol[snore$snore == "0"]
W = 0.90387, p-value = 0.005032
```

Alcohol consumption was not normally distributed in either snorers or non-snorers (Shapiro–Wilk tests, both $p < 0.05$). Therefore, a non-parametric test was used to compare the two unpaired/independent groups.

$Alcohol_{non-snorers} \rightarrow p = 6.818e - 09 < 0.05$

$Alcohol_{snorers} \rightarrow p = 0.005032 < 0.05$

```
# 4. Run the Wilcoxon rank-sum test on alcohol consumption

wilcox.test(alcohol ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data:  alcohol by snore
W = 788, p-value = 0.008594
alternative hypothesis: true location shift is not equal to 0
```

This result is statistically significant, meaning there is a difference between the two groups.

$p = 0.008594 < 0.05$

Snorers had higher alcohol consumption than non-snorers, with median values of 4 and 0, respectively. This difference was statistically significant (Wilcoxon rank-sum test, $W = 788$, $p = 0.009$), supporting the stereotype that snorers drink more.

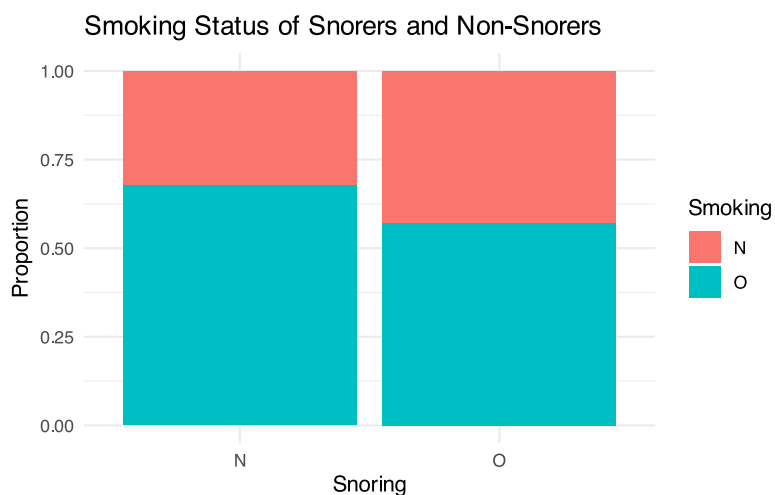
Do snorers smoke more?

The proportion of smokers is compared between snorers and non-snorers. Since smoking status is categorical (O/N), a bar plot was used rather than a boxplot.

$Smoker_{non-snorer}$ vs $Smoker_{snorer}$

```
# 1. Compare smoking distribution between snorers and non-snorers
```

```
ggplot(snore, aes(x = snore, fill = tabac)) +  
  geom_bar(position = "fill") +  
  labs(title = "Smoking Status of Snorers and Non-Snorers", x = "Snoring", y = "Proportion",  
fill = "Smoking") +  
  theme_minimal()
```



```
# 2. Create a contingency table
```

```
smoke_table <- table(snore$snore, smoke$tabac)
```

```
# a). Calculate proportions within each snoring group
```

```
prop.table(smoke_table, margin = 1) # Calculate the percentages within each row, each row adds up to 100%
```

```
      N      O  
N 0.3230769 0.6769231  
O 0.4285714 0.5714286
```

```
# b). Convert the proportions to %
```

```
prop.table(smoke_table, margin = 1) * 100
```

```
      N      O  
N 32.30769 67.69231  
O 42.85714 57.14286
```

The descriptive statistics suggest that snorers do not smoke more in this sample. About 57.1% of snorers were smokers compared with 67.7% of non-snorers, so the proportion of smokers was actually lower among snorers.

- Among non-snorers, 32.31% are non-smokers and 67.69% are smokers
- Among snorers, 42.86% are non-smokers and 57.14% are smokers

Because both variables are categorical data (0/N), a **Chi-square test of independence** instead of a parametric or non-parametric test since the data is categorical and both groups are independent (Illowsky, 2023):

```
# 3. Test whether smoking status is associated with snoring status
```

```
chisq.test(smoke_table)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: smoke_table
X-squared = 0.68872, df = 1, p-value = 0.4066
```

No statistically significant association between smoking status and snoring status. Descriptively, snorers actually smoked less often in this sample (57.1% vs 67.7%), but the Chi-square test shows that this difference is not statistically significant ($\chi^2(1) = 0.689, p = 0.407$), so the data do not support the stereotype that snorers smoke more.

$$p = 0.407 > 0.05$$

2.3.c Are Men Fatter?

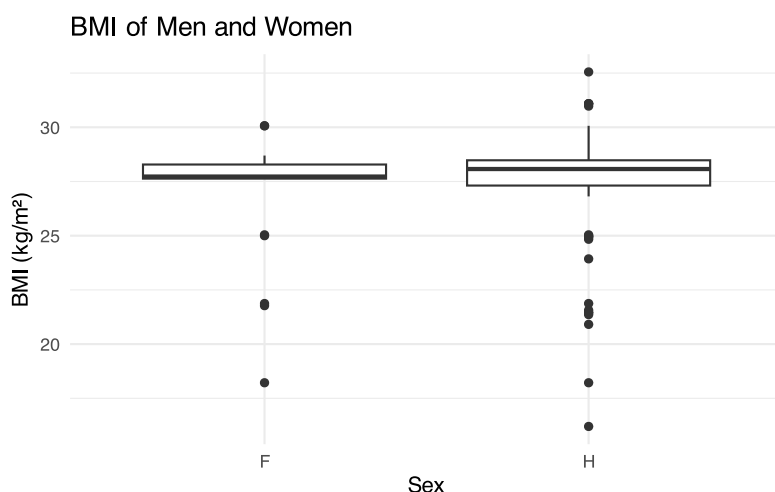
The `snore` dataset is also used to compare BMI between men and women. BMI is calculated from each participant's height and weight, and the two sex groups are treated as independent groups. The distribution of BMI is first examined before choosing the appropriate statistical test.

A box plot is first used to compare the BMI distributions of men and women before performing the statistical analysis.

BMI_{men} vs BMI_{women}

```
# 1. Make a boxplot to compare BMI between men and women
```

```
ggplot(snore, aes(x = sex, y = BMI)) +
  geom_boxplot() +
  labs(title = "BMI of Men and Women", x = "Sex", y = "BMI (kg/m²)") +
  theme_minimal()
```



```
# 2. Create a table to look at the descriptive statistics
```

```
snore %>%
```

```
group_by(sex) %>%
  summarise(mean_BMI = mean(BMI), median_BMI = median(BMI), sd_BMI = sd(BMI), n = n())
```

```
# A tibble: 2 × 5
  sex    mean_BMI median_BMI sd_BMI      n
<chr>    <dbl>      <dbl>  <dbl> <int>
1 F         27.0        27.7   2.72    25
2 H         27.3        28.1   2.85    75
```

Descriptive results suggest that men have a slightly higher BMI than women, but the difference looks small. The median BMI was 28.08 kg/m² for men and 27.73 kg/m² for women, while the mean BMI was 27.27 and 27.04 kg/m², respectively.

- Women (F): mean BMI = **27.04**, median = **27.73**, SD = **2.72**, (n=25)
- Men (H): mean BMI = **27.27**, median = **28.08**, SD = **2.85**, (n=75)

Check the normality of the data using the Shapiro-Wilk test (Ghasemi & Zahediasl, 2012)

```
# 3. Check normality of BMI in both groups using the Shapiro-Wilk test
```

```
# a) Women
shapiro.test(snore$BMI[snore$sex == "F"])
```

Shapiro-Wilk normality test

```
data:  snore$BMI[snore$sex == "F"]
W = 0.71462, p-value = 1.154e-05
```

```
# b) Men
shapiro.test(snore$BMI[snore$sex == "H"])
```

Shapiro-Wilk normality test

```
data:  snore$BMI[snore$sex == "H"]
W = 0.78184, p-value = 3.456e-09
```

BMI was not normally distributed in either women or men (Shapiro–Wilk tests, both $p < 0.05$). Therefore, a non-parametric test was used to compare BMI between the two independent groups.

- Women: $p = 1.154e - 05 < 0.05$
- Men: $p = 3.456e - 09 < 0.05$

```
# 4. Run the Wilcoxon rank-sum test on BMI
```

```
wilcox.test(BMI ~ sex, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data:  BMI by sex
```

```
W = 850, p-value = 0.4884
alternative hypothesis: true location shift is not equal to 0
```

No statistically significant difference in BMI between men and women.

- $p = 0.488 > 0.05$

Men had a slightly higher median BMI than women (28.08 vs 27.73 kg/m²), but the difference was not statistically significant (Wilcoxon rank-sum test, $W = 850$, $p = 0.488$). Therefore, the data do not support the stereotype that men are fatter.

2.3.d Do Women Smoke Less?

The `snore` dataset is used to compare smoking status between women and men. Since both sex and smoking status are categorical variables, the analysis focuses on comparing the proportion of smokers in each group.

2.3.d.a Comparison Variables

Both `sex` and `smoking` are categorical variables. Therefore, the proportion of smokers is compared between women and men.

$Smokers_{men}$ vs $Smokers_{women}$

2.3.d.b Descriptive Analysis

```
# 1. Make a proportional bar plot to compare smoking status between women and men
```

```
ggplot(snore, aes(x = sex, fill = tabac)) +
  geom_bar(position = "fill") +
  labs(title = "Smoking Status of Women and Men", x = "Sex", y = "Proportion",
       fill = "Smoking") +
  theme_minimal()
```



```
# 2. Create a contingency table for sex and smoking status
```

```
sex_smoke_table <- table(snore$sex, snore$tabac)
```

```
# a) Calculate proportions within each sex
```

```
prop.table(sex_smoke_table, margin = 1)
```

N O

```
F 0.60 0.40
H 0.28 0.72
```

```
# b) Convert the proportions to percentages
```

```
prop.table(sex_smoke_table, margin = 1) * 100
```

```
      N  O
F 60 40
H 28 72
```

A lower proportion of women were smokers compared with men (40% vs 72%). Descriptively, this supports the stereotype that women smoke less.

- **Women (F):** 40% smokers, 60% non-smokers
- **Men (H):** 72% smokers, 28% non-smokers

2.3.d.c Statistical Test Selection

Both `sex` and `tabac` are categorical variables, and the observations are independent because each participant belongs to only one sex and one smoking-status category.

A `Chi-square test` of independence can therefore be used to test whether smoking status is associated with sex.

For the Chi-square test, the expected cell counts must be reasonably large. If the expected counts are all around 5 or more, the Chi-square test is appropriate (Illowsky, 2023).

```
# 3. Test whether smoking status is associated with sex
```

```
sex_smoke_chi <- chisq.test(sex_smoke_table)
```

```
sex_smoke_chi
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: sex_smoke_table
X-squared = 7.0023, df = 1, p-value = 0.00814
```

```
# Check expected counts for the Chi-square test
sex_smoke_chi$expected
```

```
      N  O
F  9 16
H 27 48
```

The expected counts were sufficiently large, so the assumptions of the Chi-square test were considered satisfied.

2.3.d.d Conclusion

There is significant association between sex and smoking status.

- $p = 0.008 < 0.05$

A lower proportion of women were smokers compared with men (40% vs 72%). This difference was statistically significant ($\chi^2(1) = 7.00, p = 0.008$), supporting the stereotype that women smoke less in this sample.

2.3.e Are There Any Correlations Between Variables?

From a more general perspective: are there any correlations between variables?

We are no longer comparing two groups. We want to see whether the numerical variables tend to change together.

- **Numerical variables:** age, weight, height, alcohol
- **Categorical variables:** snore, gender, smoking

Categorical data should not be put in a standard correlation matrix since they are not numerical quantities. Based on the previous analysis above, some numerical variables were not normally distributed, therefore **Spearman correlation** can be used to see whether two variables generally increase or decrease together (Schober et al., 2018). Spearman is used instead of Pearson because Spearman does not require normality (Schober et al., 2018).

1. A dataset must be created for only the numerical variables.
2. The Spearman correlation matrix is calculated.
3. The interesting correlations are then tested to see if they are statistically significant or not.

```
# 1. Select the numerical variables for the correlation analysis

snore_numeric <- snore %>% select(age, weight, height, alcohol)

head(snore_numeric)
```

```
# A tibble: 6 × 4
  age weight height alcohol
<dbl> <dbl> <dbl> <dbl>
1    47     71   158      0
2    56     58   164      7
3    46    116   208      3
4    70     96   186      3
5    51     91   195      2
6    46     98   188      0
```

```
# 2. Calculate Spearman correlations between all numerical variables

correlation_matrix <- cor(snore_numeric, method = "spearman", use = "complete.obs")

correlation_matrix
```

```
      age      weight      height      alcohol
age  1.000000000 -0.01455125 0.008072461 -0.10748796
weight -0.014551254 1.00000000 0.930383489 0.08568687
height 0.008072461 0.93038349 1.000000000 0.12539769
alcohol -0.107487956 0.08568687 0.125397694 1.00000000
```

The main relationship is between weight and height. It is a very strong positive correlation. Taller participants generally tend to weigh more. All other correlations were weak or close to zero.

- $r_s = 0.930$
- 1.0 = variables are perfectly correlated with each other

Now the correlation between weight and height can be tested to see whether it is statistically significant or not. **Spearman correlation** is used to test the significance. Spearman correlation was used because some numerical

variables were not normally distributed as mentioned previously, making a non-parametric correlation more appropriate (Schober et al., 2018).

```
# 3. Test the correlation between height and weight

cor.test(snore$height, snore$weight, method = "spearman", exact = FALSE)
```

```
Spearman's rank correlation rho

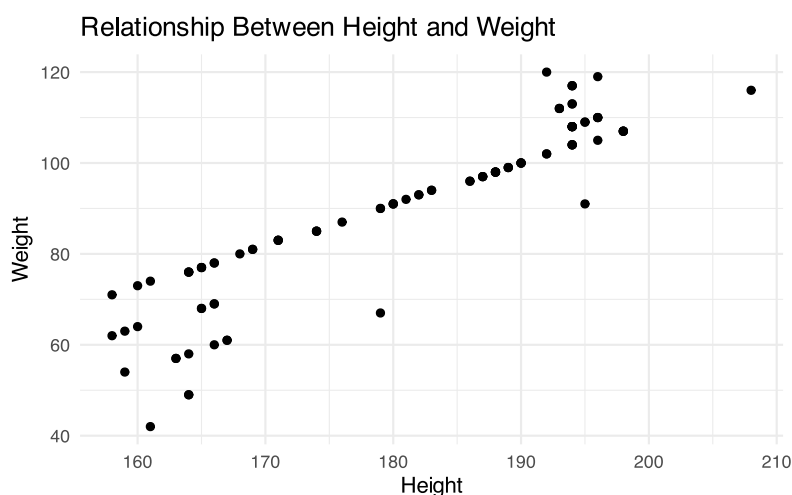
data:  snore$height and snore$weight
S = 11602, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.9303835
```

There is a very strong positive and statistically significant association between height and weight: as height increases, weight generally increases as well. $r_s = 0.930$

- $p < 2.2e - 16$

```
# 4. Visualize the relationship between height and weight

ggplot(snore, aes(x = height, y = weight)) +
  geom_point() +
  labs(title = "Relationship Between Height and Weight", x = "Height", y = "Weight") +
  theme_minimal()
```



Height and weight showed a very strong positive correlation ($r_s = 0.930$, $p < 2.2e - 16$), indicating that taller participants generally had greater body weight. The other numerical variables showed only weak or negligible correlations.

2.4 Beyond the scope

The statistical methods used in this assignment included:

Methods	Description
Descriptive statistics	Summarized the data using mean, median, SD, percent-ages, etc.
Boxplots / scatterplots	Visualized distributions and relationships

Methods	Description
Paired t-test	Compared two related measurements, such as before and after, when the differences were approximately normally distributed (Kim, 2015).
Wilcoxon rank-sum test	Compared two independent groups when data were non-normal (Whitley & Ball, 2002)
Chi-square test	Tested relationships between categorical variables (Illowsky, 2023)
Spearman correlation	Tested whether two numerical variables were related when the data were non-normally distributed (Schober et al., 2018)

The more advanced methods mentioned in the epilogue are different because they can work with several variables at the same time:

More advanced methods	What they could do
Linear regression	Use several variables to predict a numerical value
Logistic regression	Use several variables to predict a category/probability, such as snorer vs non-snorer
Clustering	Automatically find groups of similar individuals
PCA	Combine many related variables into fewer variables to simplify the dataset

- So far, the analysis has mainly used basic statistical methods to compare variables or study relationships between them, such as t-tests, Wilcoxon tests, Chi-square tests and correlations.
- More advanced data-mining methods could go further by analysing several variables at the same time. For example, age, weight, height, alcohol consumption, sex and smoking status could potentially be combined using logistic regression to predict whether a person snores. Linear regression could be used when the variable being predicted is numerical instead of categorical.
- Other methods can be used for different purposes. Clustering could identify groups of people with similar characteristics without defining the groups beforehand, while PCA could simplify a dataset containing many related variables by reducing them to a smaller number of combined variables.
- These methods could reveal patterns that are difficult to see when variables are studied separately. However, the research question should always come first, because finding a statistical pattern does not automatically mean that the pattern is useful or meaningful.

3 Data Sources

Illowsky, B., & Dean, S. (2023). *Introductory Statistics 2e*. OpenStax. <https://openstax.org/books/introductory-statistics-2e/pages/6-introduction>

Vu, J., & Harrington, D. (2020). *Introductory statistics for the life and biomedical sciences* (1st ed.). OpenIntro. biostat statistics textbook

4 Scientific References

Burgos-Sanchez, C., Jones, N. N., Avillion, M., Gibson, S. J., Patel, J. A., Neighbors, J., Zaghi, S., & Camacho, M. (2020). Impact of alcohol consumption on snoring and sleep apnea: A systematic review and meta-analysis. *Otolaryngology–Head and Neck Surgery*, 163(6), 1078–1086. <https://doi.org/10.1177/0194599820931087>

Ghasemi, A., & Zahediasl, S. (2012). Normality tests for statistical analysis: A guide for non-statisticians. *International Journal of Endocrinology and Metabolism*, 10(2), 486–489. <https://doi.org/10.5812/ijem.3505>

Kim, T. K. (2015). T test as a parametric statistic. *Korean Journal of Anesthesiology*, 68(6), 540–546. <https://doi.org/10.4097/kjae.2015.68.6.540>

Schober, P., Boer, C., & Schwarte, L. A. (2018). Correlation coefficients: Appropriate use and interpretation. *Anesthesia & Analgesia*, 126(5), 1763–1768. <https://doi.org/10.1213/ANE.0000000000002864>

Whitley, E., & Ball, J. (2002). Statistics review 6: Nonparametric methods. *Critical Care*, 6(6), 509–513. <https://doi.org/10.1186/cc1820>

5 Final Grading Checklist

This checklist was used during the pull request review process and again during the final review of the complete assignment. It helped us check that each section met the required criteria before merging and that all submission requirements were completed before the final PDF was submitted.

- ☒ Authors and date are included in the header
- ☒ Public GitHub repository is linked
- ☒ Team Git workflow strategy is explained
- ☒ Challenges encountered and lessons learned are discussed
- ☒ Complete code and answers to all statistical-test questions are included
- ☒ Data sources are provided with URLs
- ☒ Scientific references are provided with DOIs where available
- ☒ README.md is complete
- ☒ LICENSE is included
- ☒ Git history contains meaningful commit messages
- ☒ At least one commit was made per question
- ☒ Branches were used appropriately
- ☒ Main branch is up to date
- ☒ Final PDF renders correctly